

AI GENERALIST ACCELERATOR PROGRAM

AI AUTOMATIONS AND AGENTS

USE CASE - EXPENSE TRACKER

Demo Video for Reference:

<https://www.loom.com/share/7b2a93e36a5f4e0c972db749be0c33c7?t=16>

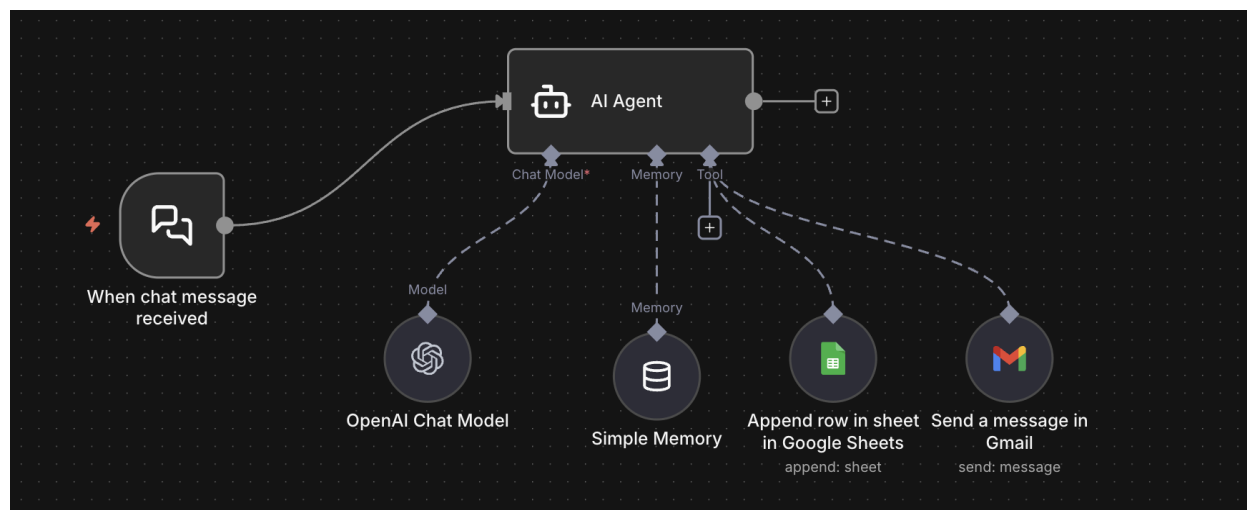
Mastering n8n: A Use-Case Driven Guide

This guide will walk you through building a powerful, AI-driven expense tracker from the ground up. We will start with a simple automation and progressively add layers of intelligence, culminating in a sophisticated AI agent that can reason and act on its own.

API KEY: [Generate one](#)

Use Case 1: Building a Simple Automation: The Basic Expense Tracker

Your Final Workflow Will Look Like This



Pre-setup: create your Google Sheet

Step 1 : Create your google sheet

Step 2 : Name The Sheet as "Expense Tracker"

Step 3 : Add Column Headers as below

- Type of Expense
- Category
- Amount
- Remaining Balance

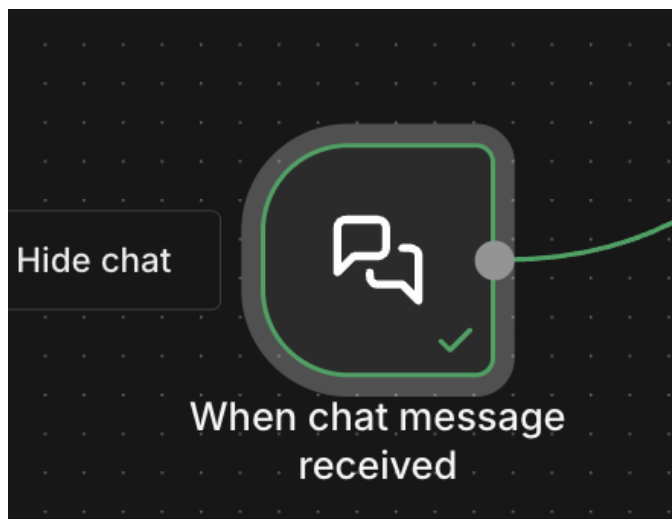
Step-by-step build (node by node)

Step 0 : Create a New Workflow

Step 1 : Add trigger node

Node type: [Chat Trigger](#)

Purpose: Accept user expense input.

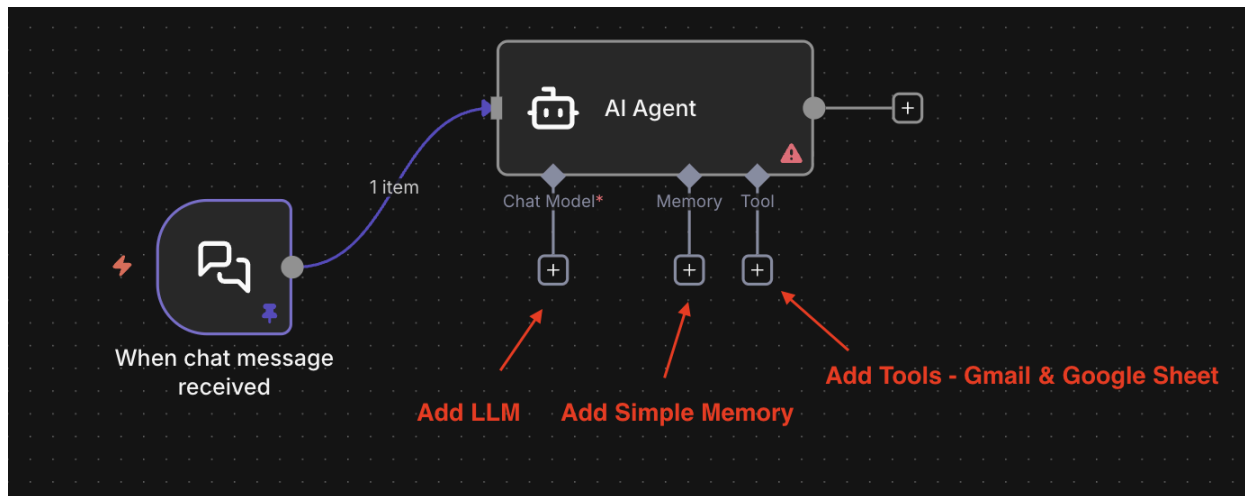


- Give a sample input "shopping spent 5000" in the chat window

Step 2 : Add AI Agent node

- Node type: **AI Agent**
- Leave it as it is for now
- We will come back to update the prompt once all nodes are connected

In the next step you add the brain , memory and Tools to the agent

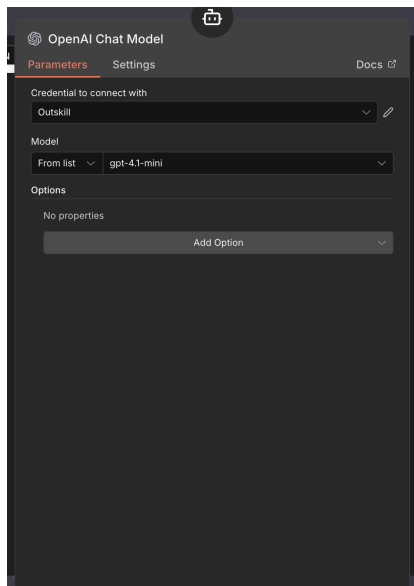


Step 3 : Add OpenAI Chat Model node (Brain of the Agent)

Node type: **Open AI Chat Model**

- Model: gpt-4.1-mini.

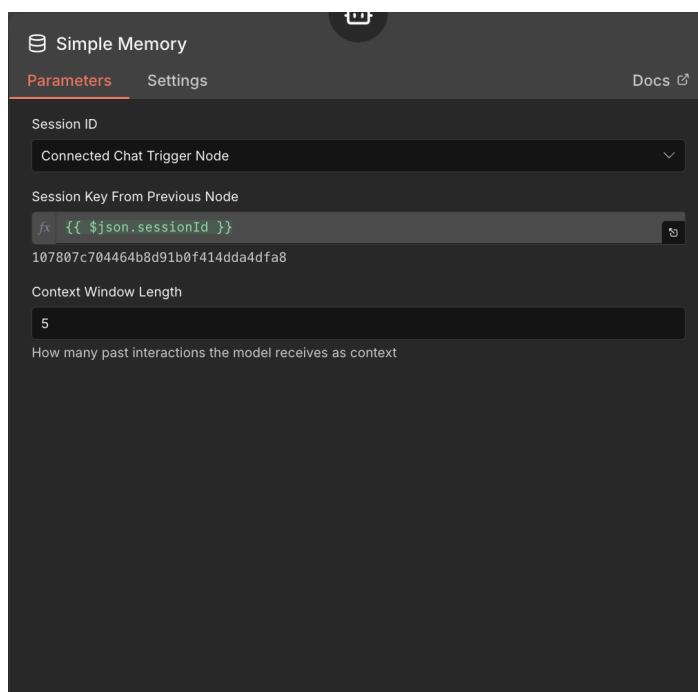
- Connect as the agent brain.



Step 4 : Add Simple Memory node

Node type: **Simple Memory**.

Purpose: keep last message for continuity.



- Context window length: 5.

Step 5 : Add Google Sheets append row node (backup)

Node type: [Google Sheets Tool](#).

The screenshot shows the configuration interface for the 'Append row in sheet in Google Sheets' node. The interface is divided into 'Parameters' and 'Settings' tabs. The 'Parameters' tab is active, showing various configuration options. At the top right, there is an 'Execute step' button. The configuration options include:

- Credential to connect with:** Uthappamdk
- Tool Description:** Set Automatically
- Resource:** Sheet Within Document
- Operation:** Append Row
- Document:** From list, Expense Tracker
- Sheet:** From list, Sheet1
- Mapping Column Mode:** Map Each Column Manually
- Values to Send:** A list of columns to be appended, each with a dropdown menu and a description field. The columns are: Type of Expense, Category, Amount, and Remaning Balance. Each column is currently set to 'Defined automatically by the model'.

- **Operation:** Append Row.
- **Document:** use your Expense Tracker sheet id.
- **Sheet:** map the sub sheet inside expense tracker
- **Columns mapping:** let the AI fill Type of Expense, Category, Amount, Remaning Balance.
- Ensure credential uses the same Google account.

Step 6 : Add Gmail Tool

Node type: **Gmail Tool**

Purpose : Use when the agent decides balance is zero or negative.

The screenshot shows the configuration interface for the 'Send a message in Gmail' tool. The interface is dark-themed and includes a title bar with the tool name and an 'Execute step' button. Below the title bar are tabs for 'Parameters' (selected) and 'Settings', and a 'Docs' link. The configuration fields are as follows:

- Credential to connect with:** A dropdown menu showing 'Gmail account 2' with an edit icon.
- Tool Description:** A dropdown menu showing 'Set Automatically'.
- Resource:** A dropdown menu showing 'Message'.
- Operation:** A dropdown menu showing 'Send'.
- To:** A text input field containing 'uthappamdk@gmail.com' with an add icon.
- Subject:** A text input field containing '+ Defined automatically by the model' with a close icon.
- + Add a description:** A button to add a description.
- Email Type:** A dropdown menu showing 'HTML'.
- Message:** A text input field containing '+ Defined automatically by the model' with a close icon.
- + Add a description:** A button to add a description.
- Options:** A section containing a toggle switch for 'Append n8n Attribution' which is currently turned on, and an 'Add option' button.

- **Tool Description** : Set Automatically
- **Resource** : Message
- **Operation** : Send
- **To** : Email Address
- let the **AI fill** Subject and Message fields
- Set append Attribution to true/false.

Step 7 : Configure the agent prompt

- Open Ai Agent Node
- Go to Options
- Choose System and click on expressions
- Paste the below system prompt (read to know the logic)

System Message :

You are a expense manager, analyse the input and segregate the output into four elements

Remaining balance is a running total, if the expense is credit, add it to the balance, else subtract from the balance

1. Type of expense (Credit/Debit)
2. Category
3. Amount
4. Remaining Balance

After every calculation, use google sheet tool to append the row.

if at any point, the balance is zero or negative, send a mail that says I'm bankrupt officially.
Use gmail send tool

Step 8 :Test steps

1. Submit sample messages:
 - a. Salary received 50000
 - b. Spent 2000 on groceriesCheck sheet for new rows and updated balance.
2. Check sheet for new rows and updated balance.
3. Force balance to ≤ 0 to test Gmail alert

Additional Resource

Link to Loom:

<https://www.loom.com/share/7b2a93e36a5f4e0c972db749be0c33c7?t=16>

Summary of Loom Recording

1. Introduction to N8N:

- Welcome to the N8N basics overview.
- Quick recap of the fundamentals of N8N.

2. Dashboard Overview [0:16](#)

- Upon logging in, you see the dashboard.
- Displays workflows created in the past.
- Shows credentials for services like Gmail, Google Sheets, etc.
- Execution history of workflows is available.

3. Creating a New Workflow [0:44](#)

- A new workflow is created in the class.
- Introduction to different types of nodes.

4. Understanding Trigger Nodes [0:54](#)

- Start with a trigger node (e.g., chat message).
- Demonstration of running the trigger node.

5. Inputting Data [1:44](#)

- Input format example: 'debit shopping 5000'.
- Explanation of how to store transactions in Google Sheets.

6. Activating the Workflow [2:33](#)

- Instructions on how to activate the workflow.
- Connecting input to Google Sheets.

7. Handling Refresh Tokens [3:30](#)

- Explanation of refresh tokens and the re-authentication process.

8. Testing the Workflow [4:20](#)

- Demonstration of reposting messages to test the workflow.

9. Splitting Input Data [5:03](#)

- Using a delimiter to split the input into an array.
- Explanation of array indexing (0, 1, 2) for data extraction.

10. Executing the Workflow [6:18](#)

- Execution of the workflow to see data populated in Google Sheets.

11. Adding Intelligence with LLM [7:20](#)

- Introduction of LLM to enhance workflow intelligence.
- Connecting chat message output to the OpenAI LLM node.

12. Setting Up OpenAI API Key [8:20](#)

- Instructions on generating and using an OpenAI API key.

13. Configuring LLM Prompts [9:54](#)

- Explanation of prompts for LLM to understand expense management.

14. Extracting Output from LLM [11:32](#)

- Connecting LLM output to Google Sheets for dynamic data insertion.

15. Implementing AI Agent [15:14](#)

- Introduction of an AI agent for advanced automation.

- Explanation of memory and modular approach.

16. Final Testing of the Workflow [20:05](#)

- Testing the complete workflow with various inputs.
- Demonstration of email alerts for specific conditions.

17. Conclusion and Future Steps [22:52](#)

- Summary of traditional vs AI automation.
- Future learning on building singular agents and multi-agent systems.